

FACULTY OF ENGINEERING

Test Strength of 3D Printed Gears for Use in Pulley System Gearbox

Checkpoint 1 Report

Module: MCTR327

Assignment: Checkpoint 1

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Submitted in pursuit of the degree

BACHELOR OF ENGINEERING

In

MECHATRONIC ENGINEERING

North-West University Potchefstroom Campus

19 September 2026

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0.1 Introduction

This report documents the testing of FDM 3D printed gears to determine their suitability in the construction of a gearbox for a gear pulley system.

0.2 Experimental Setup

The following equipment was used in the experiment:

- Load Cell
- Instrotech Model 4004 Load Cell Interpreter
- FDM 3D Printer
- PLA and PETG filament calibrated to filament roll
- Test Rig
- Jupyter and Latex

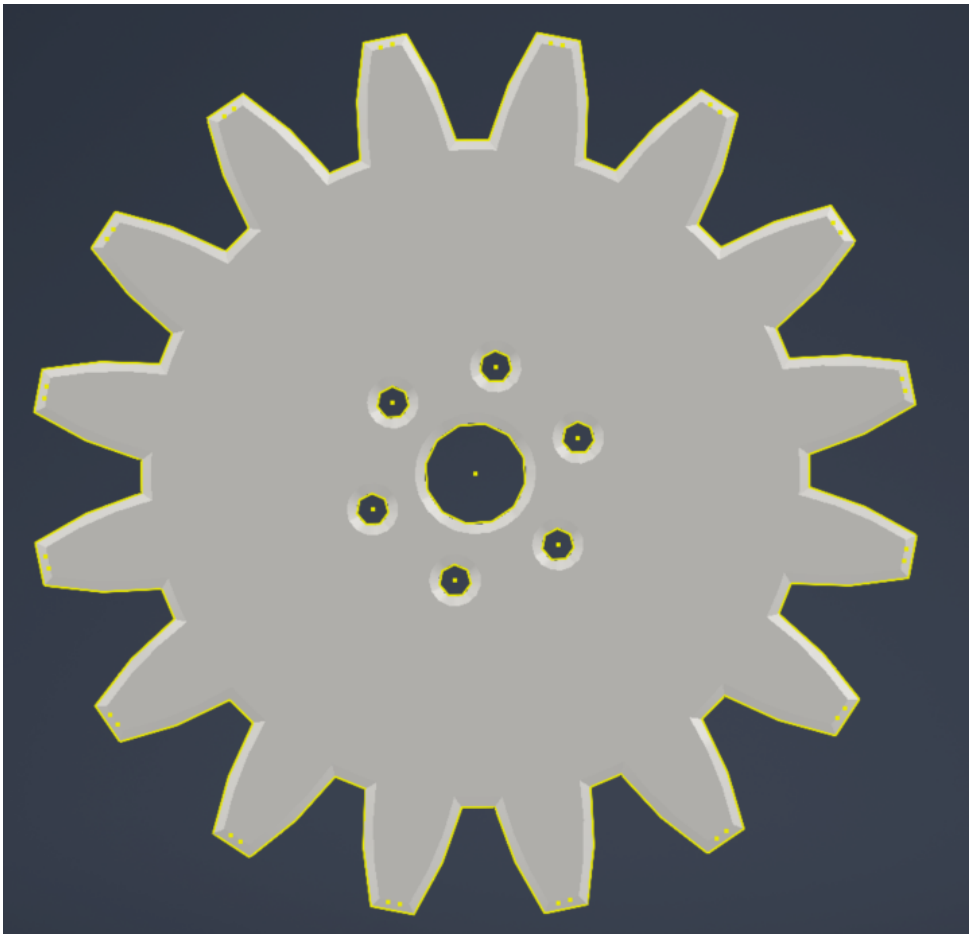


Figure 1. *Gear Design that was Tested*

A gear was designed to interface with the test rig in figure [2]. This rig uses a module 5 gear of 19mm diameter with 16 teeth and a 10mm diameter shaft hole. It has been designed with a 1mm chamfer to prevent elephant's foot while printing. It has 6 3.25mm holes for mounting

screws to the flanges of the test rig. The printer has a tolerance of 0.2mm for printing in PLA and 0.3mm for PETG [1]. The designed gear can be seen in figure [1].

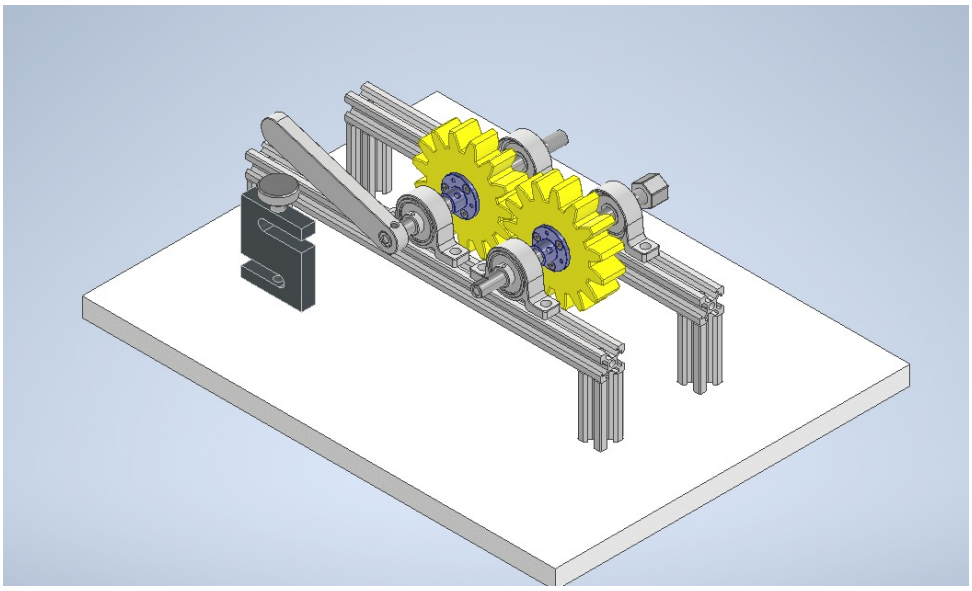


Figure 2. *CAD Design of Rig Used to Test Gear Strength*

One gear was printed with PLA and the other with PETG. Each roll was calibrated beforehand and the relevant settings were applied. The gears were set to print with the specifications in table [1]. Five layers all round are used for minimum material and maximum strength. A module five gear of this size has larger teeth than normal and if 3D printed it would have been best to print them with enough walls for solid gear teeth. This would however have increased material useage and print time with about 15 walls needed to completely fill the teeth. To compensate for this the gyroid infill setting was used to provide enough strength and general infill so that each tooth has the greatest chance of having the same strength compared to other infill settings [2]. The brim setting was used to ensure bed adhesion [4]. The printed gears are shown in figure [3].

Table 1. *Print Settings of printed gears*

Parameter	**Value**
Wall Number	Value
Bottom Shell Number	5
Top Shell Number	5
Infill Type	Gyroid
Infill %	15 %
Brim	Outer Brim Only

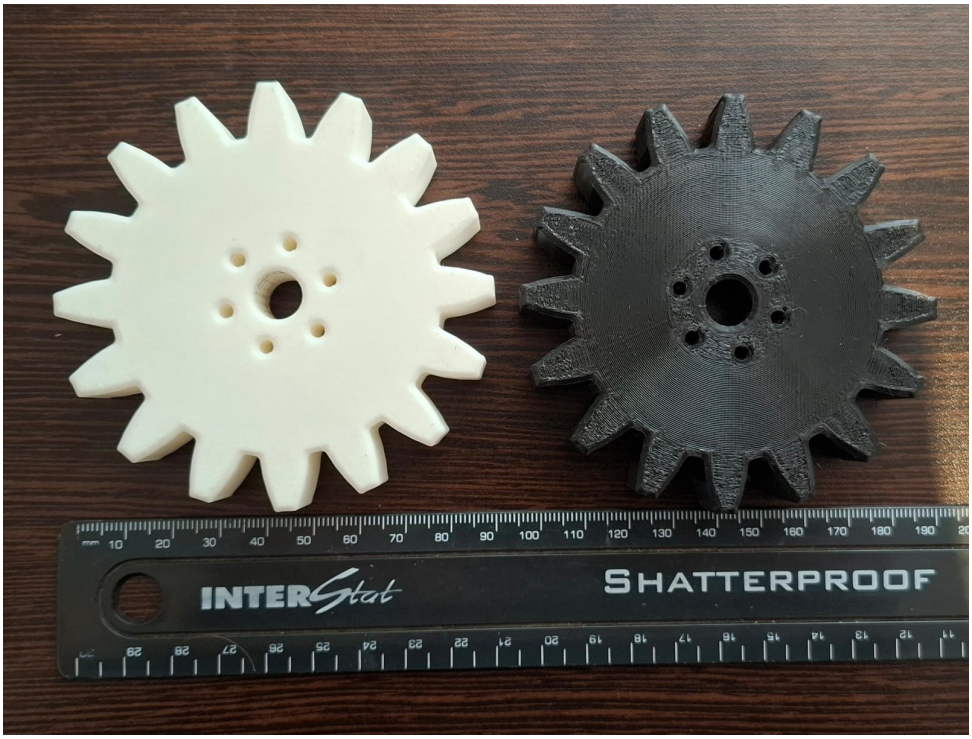


Figure 3. *Image of the 2 Printed Gears. (White = PLA; Black = PETG)*

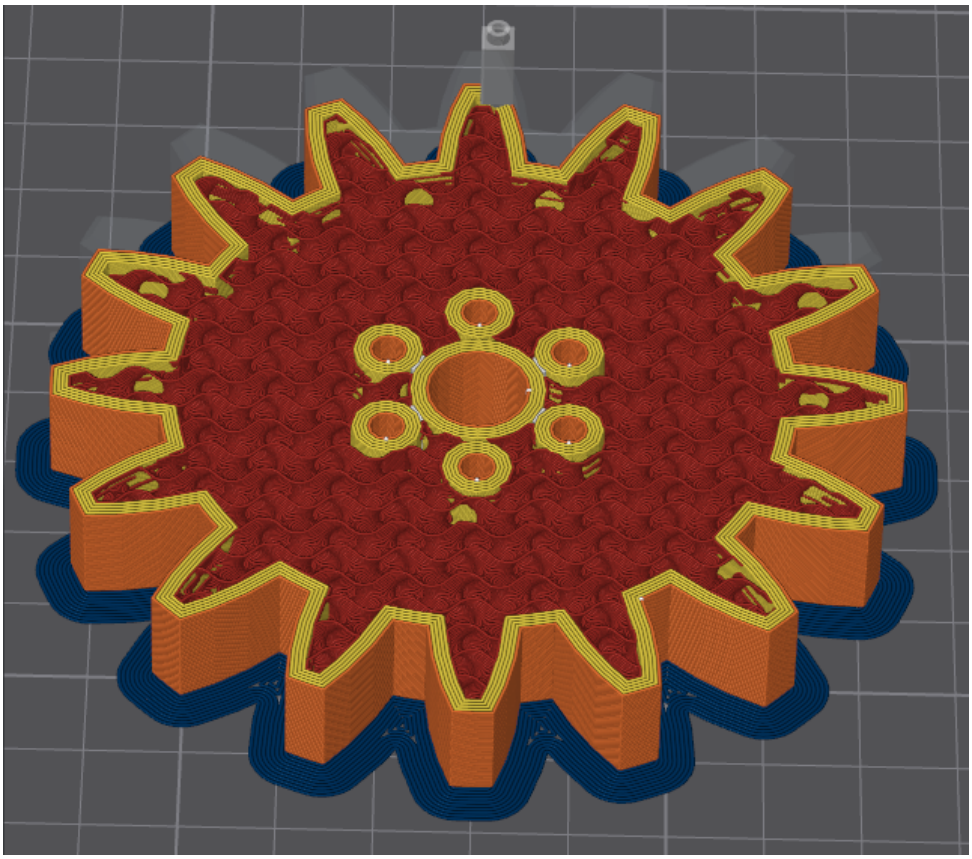


Figure 4. *Display of the internal slicer*

0.3 Method

The gears are to interface with the provided gear mounted on the test rig. A wrench would be attached to the one arm of the rig and turned to apply torque to the gears in the rig and subsequently to the arm applying the load to the load cell. The program interfacing with the Instrotech Model 4004 is set to start recording and the torque is applied until failure or the maximum load of 100 kg is applied to the load cell. The recorded results are exported as a .csv file and processed [5].

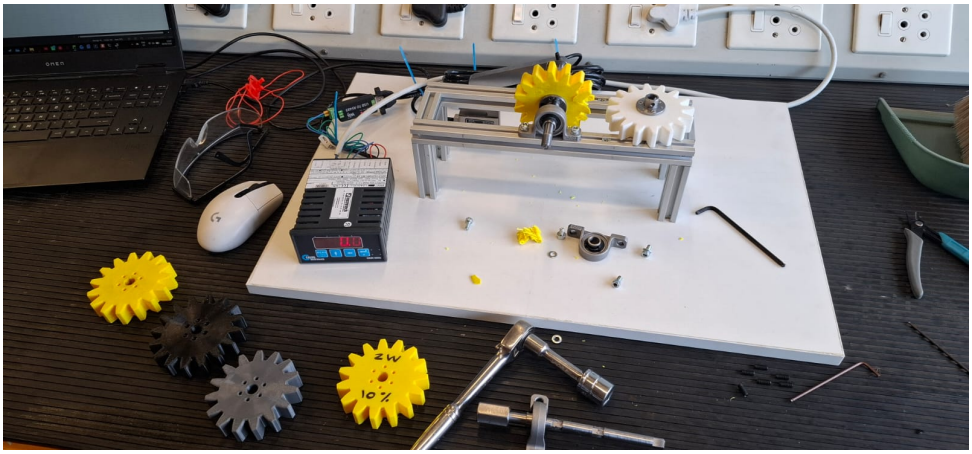


Figure 5. *Rig Used to Test Gear Strength*

To determine the results desired from the experiment, the following calculations and assumptions were made. A drum that has a height of 60mm and a diameter of 50mm was chosen. These were chosen so that the number of cable or rope can fit into only 1 wrap on the drum if the stroke is to be 1600mm if the rope were assumed to be 5mm in diameter. A mass of 30kg was assigned as the load of the drum at any one time to include the 20kg load with the equipment that might hang below it along with overhead. If the nominal speed of the motor is acheived at a speed of 1360 rpm with power of 0.18kW, then the gears need to withstand approximately 11,036Nm [1] if one is to withstand the maximum torque of the drum at any one time. It is unlikely that this top speed will be achieved by the motor due to the VFD limitations so these specifications will be prone to change [.1].

$$\tau_{\text{Drum}} = \text{Load} \times 9.81 \times \frac{\varnothing_{\text{Drum}}}{2 \times 1000} \times K_{\text{Serving Factor}}$$

Using the maximum load from the results, the torque applied to the gear can be determined and compared to the required torque. The test rig arm pressing onto the load cell is 100mm long[12].

$$\tau = F \times d$$

0.4 Results

The results from the rig are rather the strength of the rig rather than that of the gears tested. The yellow gears in figure [5] had failed before the loaded white and black gears. The PETG gear was used as the test gear agains the PLA gear. In this configuration the shaft failed as can be seen in figure [6]. The results are however still usable.

From the maximum load result of 16.8kg from the second PLA test, it can be calculated that the torque is 16.68Nm [12].

$$\tau = F \times d$$

$$\tau = 16.8 \times 9.81 \times 0.01$$

$$\tau = 16.68\,\mathrm{Nm}$$



Figure 6. *Rig Failure During Test*

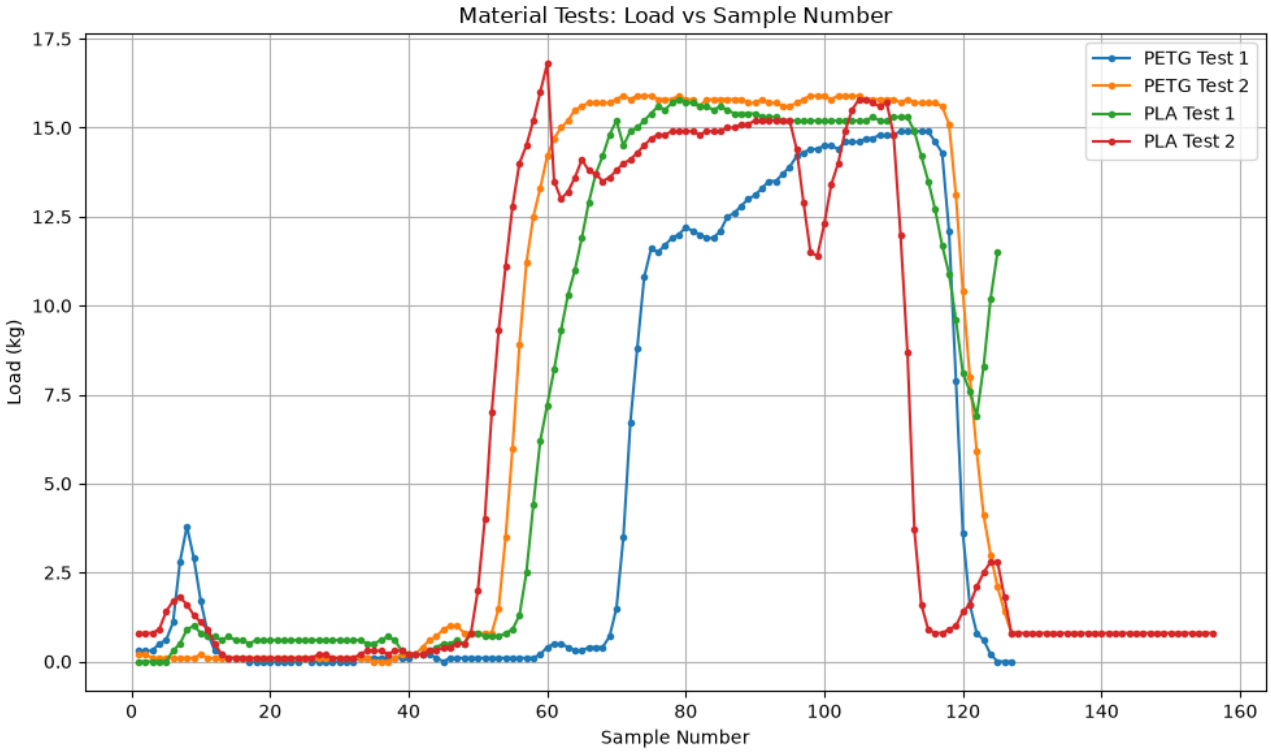


Figure 7. *Results of the gears tested*

0.5 Discussion

From the results seen in figure [7], it can be seen that the maximum load carried by the rig gears before gear or shaft failure of the rig was around 17kg for the second PLA gear test. Both the PLA and PETG gears seen in figure [3] had not failed, implying they will be able to withstand even greater loads.

The mounting options for the gears to the shafts will need to be tested as well [2] [3]. The gears used in the gearbox will also be of a smaller module, and thickness to be adhere to space limitations on the rig.

0.6 Conclusion

The gears tested will be able to bear the load necessary if one gear were to be bear all the torque of the entire load. This has been concluded as unlikely and since the gears of the test rig failed, the implication is that the tested gears will be able to withstand even greater loads. The poorly printed gears withstood 16.68Nm of torque while the requirement is only 11Nm The final gears to be used in the gearbox will however also be tested before final implementation to also test whether the gear mounting option chosen is sufficient.

References

[1] Bkingan, “Tensile testing results of toner plastics 3d filament - toner plastics,” 2026. [Online]. Available: https://toner-plastics.com/tesile-testing-results-of-toner-plastics-3d-filament/

[2] Projects, “How to 3d print gears that actually work - youtube,” 2026. [Online]. Available: https://www.youtube.com/watch?v=MOYUi-ohu0M

[3] R. G. Budynas and J. K. Nisbett, *Shigley’s Mechanical Engineering Design*, 11th. New York, NY: McGraw-Hill Education, 2019, ISBN: 9780073398211.

.1 Appendix A: Excel Tables for Calculations

Table 2. *Design main specifications*

Design Main Specifications		
Max Load	30	kg
Stroke	1600	mm
	80	mm/s

Table 3. *Gear specifications*

Gear Specs			
Parameter	**Value**	**Unit**	**Comment**
Module	2	ul	
Number_of_Teeth	30	ul	
Pressure_Angle	25	deg	
Gear_Height	15	mm	
Pitch_Circle	60	mm	
Addendum_Circle	64	mm	
Root_Circle	55	mm	
Clearance_Circle	56	mm	
Base_Circle	54.37847	mm	
Reduction Ratio	44.5059		
Per Stage work	6.671274		
Np	11		
Ng	250		
m	22.72727		

Table 4. *Drum specifications*

Drum Specs		
Line Speed	88	mm/s
Diameter	50	mm
Height	60	mm
Starting Serving Factor	1,5	
Circumference	157.0796	mm
Loops of Circ fitted	12	
Rope Length in 1 layer	1884.956	
Rpm	30.55775	
Torque	11.03625	Nm

Table 5. *Rope specifications*

Rope Specs		
Diameter	5	mm
Length	1600	mm
Wraps	10.18592	

Table 6. *Three-phase motor specifications*

3-Phase Motor Specs		
Power	0.18	kW
Speed	1360	rpm
Mass	4.83	kg
Shaft	11	mm
Keyway	4*16	mm
Rated Torque	1.263877	Nm
Starting Torque	3.791632	Nm

.1 Appendix B: Equations

$$\tau_{\mathrm{Drum}} = \mathrm{Load} \times 9.81 \times \frac{\varnothing_{\mathrm{Drum}}}{2 \times 1000} \times K_{\mathrm{Serving\ Factor}} \tag{1}$$

$$\mathrm{N_P} = \frac{2\mathrm{k}}{(1+2\mathrm{m})\sin^2\phi}\left(\mathrm{m} + \sqrt{\mathrm{m}^2 + (1+2\mathrm{m})\sin^2\phi}\right) \tag{2}$$

$$\mathrm{N_G} = \frac{{\mathrm{N_P}}^2 \sin^2\phi - 4k^2}{4k - 2N_P \sin^2\phi} \tag{3}$$

$$D_P = \mathrm{module} \times \mathrm{z} \tag{4}$$

$$\mathrm{D} = \frac{p}{\pi} \tag{5}$$

$$D = D_p + 2 \times m \tag{6}$$

$$D_r = D_p - 2 \times b \tag{7}$$

$$b = 1.25 \times m \tag{8}$$

$$D_c = D_p - 2 \times m \tag{9}$$

$$p = m \times \pi \tag{10}$$

$$D_b = D_p \cos \alpha \tag{11}$$

$$\tau = F \times d \tag{12}$$